

Understanding the Exoplanet Prediction Model

This document explains the features used by our machine learning model to predict whether a signal from space is a **Confirmed Planet** or a **False Positive**. The goal is to provide a clear understanding of the data so that anyone, from astronomers to curious enthusiasts, can effectively use and interpret the model's predictions.

What Does the Model Do?

The model is a **Voting Ensemble**, which means it intelligently combines the predictions of two powerful algorithms (Tuned XGBoost and Random Forest) to make a final, more accurate decision. It was trained on a massive dataset combining observations from two of NASA's most successful planet-hunting missions: the **Kepler Space Telescope** and the **Transiting Exoplanet Survey Satellite (TESS)**.

The model's job is to look at the data for a potential planet and classify it into one of two categories:

- **Confirmed Planet (Class 1):** The signal is very likely caused by a real planet orbiting its star.
- **False Positive (FP) (Class 0):** The signal mimics a planet but is likely caused by something else, such as a background eclipsing binary star, stellar activity (like starspots), or instrumental noise.

How Was the Data Prepared?

To build a robust model, data from Kepler and TESS were merged. This involved a few key steps:

1. **Standardizing Features:** Kepler and TESS used different names for the same measurements (e.g., Kepler's `koi_period` and TESS's `pl_orbper` both measure orbital period). We mapped these to a single, unified set of feature names.
2. **Simplifying Classes:** The original datasets had several classifications like "CANDIDATE" or "PLANET CANDIDATE". To make the model's task clear, we removed these ambiguous cases and focused only on signals that had been confidently labeled as either **Confirmed Planet** or **False Positive**.

How Well Does It Perform?

The model is highly effective and reliable, with performance that can be summarized by a few key metrics:

- **Overall Accuracy (91%):** Out of every 100 signals it examines, the model correctly classifies 91 of them on average.
- **Precision for False Positives (95%):** When the model predicts a signal is a **False Positive**, it is correct 95% of the time. This is crucial for confidently filtering out noise and not wasting time on dead ends.
- **Recall for Confirmed Planets (92%):** The model successfully identifies 92% of all the *true* planets in the dataset. This means it has a very low rate of missing actual exoplanets.

- **ROC-AUC Score (0.9705):** This is a technical measure of diagnostic ability. A score this close to 1.0 indicates that the model has an **outstanding** ability to distinguish between real planets and false positives.

Understanding the Model's Features (The Inputs)

The model uses 24 distinct features to make its predictions. These features describe the potential planet, its star, and its orbit. They are explained below in simple terms.

Core Observational Features

These features are directly related to the "transit" method—the dimming of starlight as a planet passes in front of its star.

Feature Name	Description
orbital_period	The time it takes for the planet candidate to complete one full orbit around its star. Measured in days .
transit_depth	How much the star's light dims when the planet passes in front of it. A deeper dip can mean a larger planet. Measured in parts per million (ppm) .
transit_duration	The length of time the star's light is dimmed during a transit. Measured in hours .
signal_to_noise	How clear and strong the transit signal is compared to random background noise. A higher value means a more confident detection.
num_transits	The total number of transits observed for this candidate. More transits increase the certainty that the signal is real and periodic.

Host Star Properties

These features describe the star that the candidate planet is orbiting.

Feature Name	Description
stellar_temp	The effective surface temperature of the host star. This relates to the star's color and age. Measured in Kelvin (K) .
stellar_radius	The size of the host star. Measured in multiples of our Sun's radius (e.g., a value of 2 means the star is twice as big as the Sun).
stellar_logg	The star's surface gravity. It's a measure of how dense the star is. A high value suggests a smaller, denser star.

Inferred Planetary & Orbital Properties

These are characteristics of the planet and its orbit that are calculated from the observational data.

Feature Name	Description
planet_radius	The estimated radius (size) of the planet candidate. Measured in multiples of Earth's radius (e.g., a value of 10 means it's 10 times larger than Earth).
planet_insol	The amount of radiation (light and heat) the planet receives from its star compared to what Earth receives from the Sun. Earth's value is 1.
planet_teq	The planet's "equilibrium temperature"—a calculated surface temperature assuming it has no atmosphere. This helps estimate if it's hot, cold, or temperate. Measured in Kelvin (K) .

semi_major_axis	The average distance from the planet to its star, which defines the size of its orbit. Measured in Astronomical Units (AU) , where 1 AU is the Earth-Sun distance.
impact_parameter	Describes how centrally the planet transits its star. A value of 0 means it crosses the exact center, while a value of 1 means it just grazes the star's edge.

Engineered & Technical Features

These features were created from the original data to provide more powerful insights for the model.

Feature Name	Description
depth_per_stellar_rad	Normalizes the transit depth by the star's radius. This helps the model understand the transit's significance relative to the star's size.
planet_to_star_ratio	A direct comparison of the planet's radius to the star's radius. This ratio is fundamental to understanding the transit signal.
orbital_velocity_proxy	An estimate of how fast the planet is moving, calculated from its orbital period using Kepler's Third Law of planetary motion.
temp_ratio	The ratio of the planet's equilibrium temperature to its star's temperature. This provides a scale-independent measure of the system's thermal properties.
log_orbital_period	The logarithm of the orbital period. Astronomical data can span huge ranges, and a log scale helps the model find patterns more easily.

log_transit_depth	The logarithm of the transit depth. This helps the model handle the wide variation in transit depths more effectively.
log_planet_insol	The logarithm of the planet's insolation flux.
in_habitable_zone	A simple binary flag (1 for yes, 0 for no) indicating if the planet's temperature falls within a range (200–350 K) where liquid water could potentially exist.
is_kepler	A binary flag that tells the model the data source: 1 if it's from the Kepler mission, 0 if it's from TESS. This helps the model account for the different characteristics of each telescope.
ra / dec	The celestial coordinates (Right Ascension and Declination) telling us the object's location in the sky, similar to longitude and latitude on Earth. Measured in degrees .

How to Use the Model

Using the model is a straightforward process. Follow these steps to get a prediction for your own exoplanet candidate.

Step 1: Prepare Your Input Data

You must provide a value for **each of the 24 features** listed above. The model was trained on this specific set of information, and it cannot make a prediction without all of them.

- **Format:** The data for a single candidate should be structured so that each value is clearly labeled with its corresponding feature name (e.g., in a Python dictionary, a JSON object, or a single-row CSV file).
- **Consistency:** Ensure the units and scales match the descriptions above (e.g., `orbital_period` in days, `planet_radius` in Earth radii).

For example, your input for one candidate might look like this:

```
{'orbital_period': 10.5, 'transit_depth': 850.2, 'transit_duration': 3.1, ... and so on for all 24 features}
```

Step 2: Get a Prediction from the Model

Feed your prepared input data into the model. The model will process this information and produce an output.

Step 3: Interpret the Output

The model provides two key pieces of information: a final classification and a confidence score.

1. **Final Prediction:** This is the model's "yes" or "no" answer.
 - **1:** The model classifies the signal as a **Confirmed Planet**.
 - **0:** The model classifies the signal as a **False Positive**.
2. **Probability Score:** This number, ranging from 0.0 to 1.0, tells you **how confident** the model is in its prediction.

Example Scenarios:

- **High Confidence Planet:** An output of `[Prediction: 1, Probability: 0.97]` means the model is 97% sure the candidate is a **Confirmed Planet**. This is a high-priority candidate for follow-up observation.
- **High Confidence False Positive:** An output of `[Prediction: 0, Probability: 0.99]` means the model is 99% sure the signal is a **False Positive**. This candidate can likely be deprioritized.
- **Uncertain Candidate:** An output of `[Prediction: 1, Probability: 0.55]` means the model leans towards it being a planet but has very low confidence. This candidate is ambiguous and may warrant further data collection or a closer look.

By using both the prediction and the probability score, you can make more informed decisions about which candidates are worth your time and resources.